

Finding MeBo: Delivering Reminiscence Therapy for Older Adults Using LLM-Based Voice User Interfaces

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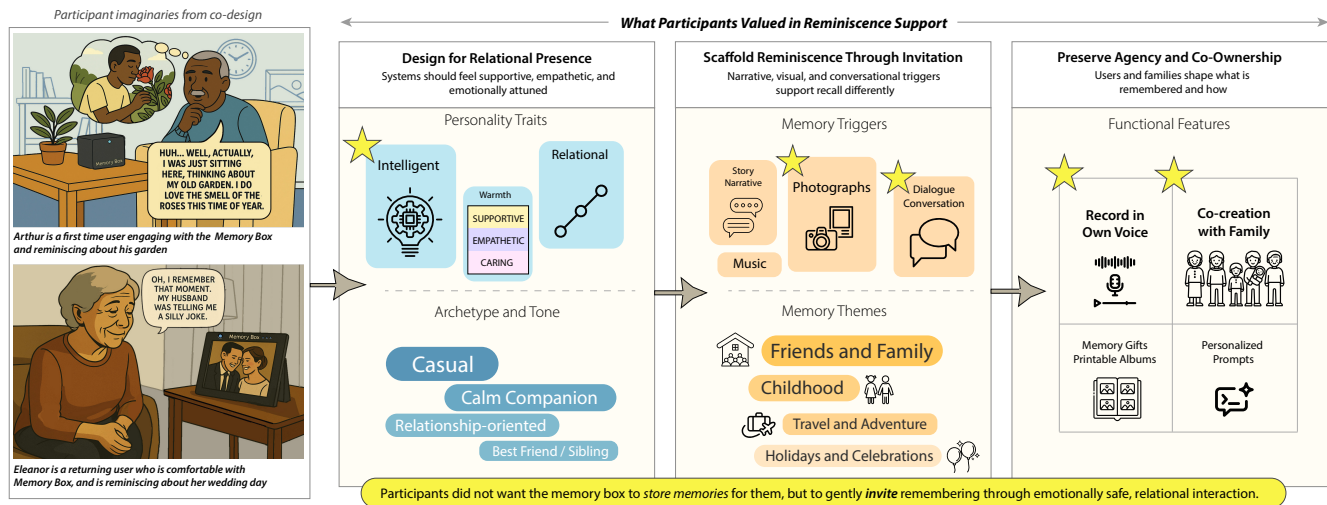


Figure 1: Design insights from co-design sessions with older adults using a conceptual voice-based *Memory Box* for reminiscence. Storyboards (left) show participant imaginaries that helped guide discussion during the workshops. Synthesized themes (right) summarize the system qualities participants valued most, including relational personality, memory triggers, and useful features. Rather than showing a complete feature set, the figure highlights design considerations that support emotionally safe, relational, and participant-driven reminiscence.

Abstract

We present a participatory co-design study with older adults (N=11) exploring how a conceptual large language model (LLM)-based voice system, *Memory Box*, can support reminiscence. Participants consistently framed *Memory Box* as a relational and emotionally consequential presence rather than a neutral or task-oriented tool, evaluating interaction in terms of attentiveness, emotional safety,

conversational continuity, and fit within existing social and intergenerational contexts. Meaningful reminiscence was grounded in relational memory themes and supported by combinations of conversational, visual, and auditory cues, with photos and music functioning as conversational anchors rather than standalone interventions. Participants further emphasized features that support co-creation, sharing, and legacy-building through voice recordings and tangible artifacts. Drawing on these findings, we articulate design guidelines that foreground relational interaction, multimodal support, and continuity over time.

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CCS Concepts

• **Human-centered computing** → Empirical studies in HCI; Natural language interfaces; Sound-based input / output.

Keywords

remembrance, conversational AI, voice interfaces, large language models, co-design

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1 Introduction

Memory plays a central role in how older adults make sense of their lives, relationships, and identities. Reminiscence therapy (RT) leverages this through sensory prompts such as photographs, music, and personal artifacts to support the recall and sharing of past experiences [6, 32]. In practice, however, the benefits of RT hinge on interactional qualities such as pacing, attentiveness, and sensitivity to emotional cues, typically provided by trained human facilitators [5, 9]. Recent advances in Large Language Models (LLMs) introduce the possibility of voice-based systems that can dynamically generate prompts, maintain conversational context, and adopt distinct personas [7, 34], yet these same capabilities complicate the facilitator role, introducing risks such as hallucinated content [12] and the potential to surface emotionally complex memories without appropriate sensitivity [10, 20]. As LLM-based voice systems increasingly occupy roles once held by human facilitators, we lack empirical understanding of how older adults perceive and evaluate these systems as sources of reminiscence support.

Prior HCI work has explored technology-mediated reminiscence primarily through systems that digitize or augment traditional memory *themes* and *triggers*, such as photographs, music, or personally meaningful life events [22, 23, 27], often positioning technology as artifact or scaffold rather than active conversational participant [13]. More recent work has begun to examine conversational agents as facilitators, showing that tone and *persona* shape engagement [15, 24], though challenges remain around pacing, speech variability, and supporting pauses during recall [3, 11]. Rather than prescribing a specific system design, this evolving landscape motivates closer attention to how such systems are understood, evaluated, and imagined by older adults themselves.

In this work, we conducted a participatory co-design study with older adults (N=11) to investigate how they reason about an LLM-based, voice-driven reminiscence concept—*Memory Box*—and what they consider essential for meaningful and emotionally safe support. Our study was guided by three research questions: (RQ1) what types of personas older adults consider suitable; (RQ2) what memory themes and triggers (e.g., photos, music, objects) they find most meaningful; and (RQ3) what functional features they identify as desirable.

Our findings reframe *Memory Box* as a relational, emotionally attuned presence, with memory cues as conversational anchors, with support emerging through trust and continuity. Design guidelines emphasizing emotional safety, multimodal dialogue, and sustained relational engagement for meaningful reminiscence.

2 Related Work and Background

Reminiscence Therapy is a psychosocial intervention that uses sensory prompts, such as photographs, music, or personal artifacts, to support the recall and sharing of past life experiences [6, 32]. RT is widely recognized as a non-pharmacological approach for improving well-being among older adults, with prior work demonstrating reductions in depressive symptoms alongside increases in life satisfaction and positive affect [5, 25, 29]. Reminiscence is also inherently social: sharing memories promotes interpersonal interaction, strengthens social bonds, and can reduce loneliness and social isolation [8, 9, 31]. In practice, however, these benefits typically depend on trained human facilitators who can structure conversations and respond sensitively to emotional cues [5, 9], raising questions about how such interactional qualities can be supported through technology.

Prior work on technology-mediated reminiscence has largely focused on digitizing or augmenting traditional memory triggers, including digital storytelling systems [22, 23], interactive photo albums [27], and immersive VR environments recreating meaningful places [16, 18]. While these approaches broaden access to multimedia cues, they often function as passive artifacts or continue to rely on human facilitation for conversational guidance and meaning-making [13]. More recent systems explore conversational agents as active facilitators, showing that conversational tone and persona shape engagement and self-disclosure [15, 24]. Voice User Interfaces are particularly promising for older adults due to their hands-free and accessible interaction style, though challenges remain around pacing, speech variability, and supporting pauses during recall [3, 11, 24].

Recent advances in LLMs further expand this design space. LLM-Based agents can dynamically generate prompts, maintain conversational context over time, and adopt distinct conversational personas [7, 34]. Multimodal LLMs additionally allow voice-based systems to reference personal visual artifacts, integrating photographs or music as conversational memory triggers [30]. At the same time, these capabilities introduce risks, including hallucinated content [12] and the possibility of surfacing emotionally complex or bitter-sweet memories without appropriate safeguards [10, 20]. Rather than prescribing a specific system design, this literature maps a space of interaction strategies, modalities, and risks that inform the *concept* of *Memory Box*. We synthesize these insights into five system-level design considerations (DCs) that guide our exploration (see Table 1).

Prior research also emphasizes the importance of involving older adults directly in shaping reminiscence technologies. Participatory and co-design studies show that older adults engage more effectively with concrete, evocative artifacts, such as storyboards or memory prompts, than with abstract feature descriptions [17, 19]. Recent work further demonstrates that digitally mediated co-design with older adults is feasible and productive when supported by care-full facilitation practices, including flexible pacing, clear turn-taking, and tangible-at-a-distance materials [1, 4, 21]. Building on this work, we highlight three co-design methodological strategies (CoMS) in Table 1 to structure our sessions, using speculative probes to surface priorities, boundaries, and sensitivities that shape how older adults reason about an LLM-Based, voice-based reminiscence

concept. Older adults' adoption of conversational and smart home technologies is shaped by privacy and control concerns, especially when data practices lack transparency [26].

3 Methods

We conducted remote participatory co-design sessions to probe how older adults interpret and respond to a voice-based, multimodal reminiscence concept, *Memory Box*, and to surface experiential constraints that shape acceptable facilitation, memory cues, and system capabilities. Our methodological choices were guided by the design considerations (DC1 to DC5) and co-design methodological strategies (CoMS1 to CoMS3) synthesized from prior work (Table 1).

Participants. Eleven older adults (6 women, 5 men) between the ages of 63 and 78 years ($M = 68.9$, $SD = 5.07$) participated in 90-minute remote co-design sessions. All were native English speakers residing in the United States. The sample was relatively highly educated: six participants held graduate degrees, two held bachelor's degrees, and three had completed some college education. Prior experience with voice-based conversational agents varied substantially; five participants reported no ownership of smart speakers or smartphone-based assistants, while the remaining participants owned at least one device, including standalone smart speakers, smartphone-based assistants, or both, with one participant reporting multiple smart speakers in addition to a smartphone assistant. Self-rated familiarity spanned the full five-point scale, ranging from not at all familiar (3 participants) and slightly familiar (1) to moderately familiar (3), very familiar (3), and extremely familiar (1). Participants were recruited through community centers, senior organizations, and word-of-mouth and received a \$30 gift card for their involvement.

Material. Found in Figure 2, low-fidelity storyboards depicted *Memory Box* scenarios to ground discussion in concrete, memory-evoking contexts (CoMS1) and surface design considerations (DC1–DC5). The Miro board supported three co-design exercises: (i) memory themes and triggers with participant-defined additions (CoMS2), (ii) persona curation via adjective cards and archetype prompts (CoMS3), and (iii) feature ideation (“I wish the Memory Box could...” with prioritization. Storyboards and Miro Boards we created have been included in the supplementary material.

Procedure. All sessions followed IRB-approved procedures and lasted 90 minutes. Sessions used secure video conferencing; facilitators presented design probes and coordinated Miro-based activities via laptop. We began with an accessible orientation and warm-up memory sharing, then storyboard exploration illustrating voice-only, multimodal, and future-oriented interaction. We then conducted three exercises on the Miro board: (1) sort/prioritize memory themes and link to sensory cues; (2) curate VUI personas and relational boundaries; (3) generate and rank desired features. Debrief captured overall impressions, emotional impact, and privacy concerns.

Data Analysis. Interviews were recorded via Microsoft Teams, transcribed, and anonymized. We conducted reflexive thematic analysis [2], synthesizing transcripts with Miro board artifacts to contextualize participant remarks within design decisions. Analysis followed an iterative, open coding process, tagging data for semantic and latent meaning across the three research questions.

Codes and themes were developed reflexively rather than from a pre-defined framework; the full codebook is provided in the supplementary material. Results are reported using standardized qualitative descriptors—a few (<20%), some (21–50%), most (51–80%), and nearly all (>80%)—to convey prevalence.

4 Findings

Synthesizing the data gathered from the co-design artifacts revealed a consistent interpretive stance across participants. *Memory Box* was seen as a relational, emotionally meaningful presence rather than a neutral tool, evaluating interactions by attentiveness, conversational continuity, and emotional safety. These framings shaped their reasoning about personas and roles (RQ1), memory themes and triggers (RQ2), and features for sustained, safe engagement (RQ3).

4.1 RQ1: Preferred Personas for Reminiscence Therapy

Emotional tone and conversational intelligence. There was strong convergence around personas characterized by emotional warmth and attentiveness. *Nearly all* participants rejected clinical, authoritative, or diagnostic framings, instead describing ideal personas as companions who support reminiscence through attentive listening and gentle prompting (“*what you’ve got here... is something that’s very much mimicking what a real life caring caregiver is doing,*” P7). Some participants associated expert or formal personas with discomfort or reduced engagement, noting that judgment or authority would inhibit interaction (“*non-judgmental’s got to go without saying. I don’t want this thing judging me,*” P7). Intelligence was valued as conversational judgment rather than factual expertise, including the ability to ask appropriate follow-up questions, pause when needed, and allow stories to unfold naturally (“*the memory box has to know to shut up and let them dance,*” P11). Most participants also welcomed light humor when used sparingly to support comfort.

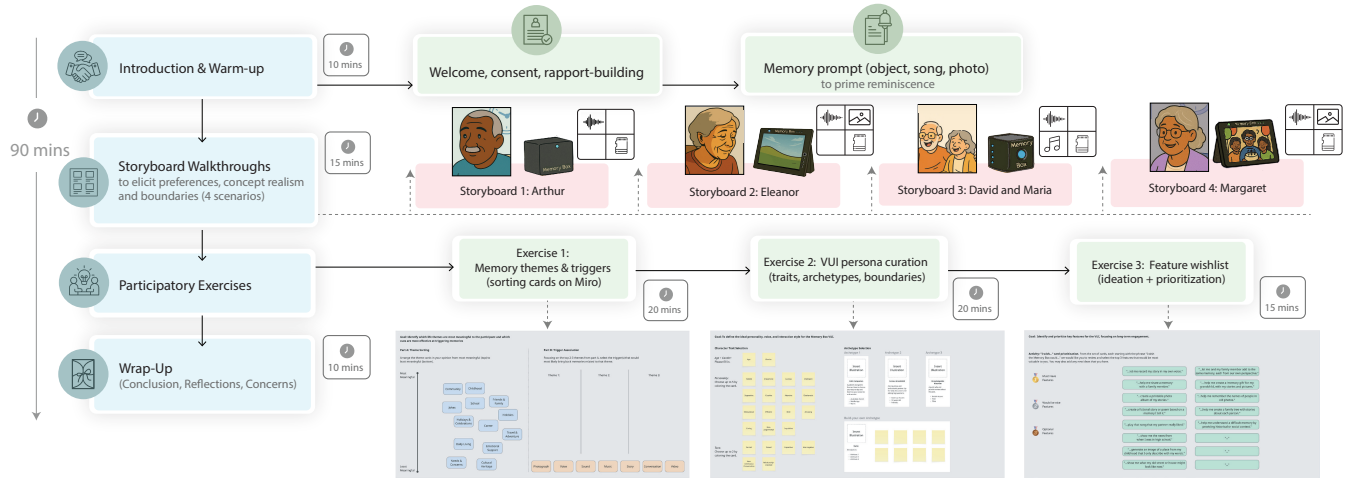
Role framing and relational orientation. Participants gravitated toward familiar and relational archetypes such as friends, siblings, or calm companions. *Nearly all* participants valued these roles for fostering trust and ease, positioning the system as someone to talk with rather than someone who directs interaction (“*I would call it like a best friend or a buddy,*” P5). Some participants emphasized conversational balance, preferring systems that listen more than they speak and avoid overly structured interaction (“*let me enjoy the memory rather than rush on to the next thing,*” P11).

4.2 RQ2: Meaningful Memory Themes and Triggers

Relational and experiential memory themes. *Nearly all* participants identified memories of friends and family as the most vivid and emotionally resonant, often emerging naturally through storytelling rather than structured recall (“*Well, I have the photo of my mother and my father... I made, like, a collage... I look at it every day,*” P9). Most highlighted childhood, travel, holidays, and celebrations for their narrative richness, while some valued community involvement and hobbies embedded in shared social contexts

Table 1: Design Considerations and Co-Design Strategies derived from Literature.

Category	Consideration	Literature-Informed Rationale
<i>System Design Considerations (DCs)</i>		
DC1	Intelligent Facilitation	Move beyond factual recall by structuring conversations with themes and providing scaffolds (nudges, summaries) to deepen reflection [7, 34].
DC2	Emotional Complexity	Detect emotional cues to gently guide users toward positive appraisals or offer clear exit paths if distress arises [10, 20].
DC3	Longitudinal Context	Maintain a user model that recalls past conversations and boundaries to build rapport and evolve as a partner [15, 24].
DC4	Multimodal Integration	Interoperate with visual (photos) and auditory (music) artifacts to allow the LLM to reference displayed content [30].
DC5	Sensory Adaptation	Accommodate sensory decline via slower TTS pacing, clear prosody, and error-tolerant command structures [11].
<i>Co-Design Methodological Strategies (CoMS)</i>		
CoMS1	Evocative Grounding	Use concrete artifacts (storyboards) rather than abstract lists, as older adults engage more readily with tangible triggers [27, 33].
CoMS2	Trigger Prioritization	Employ card sorting to rank themes/triggers, acknowledging that trigger-task fit shapes reminiscence quality [33].
CoMS3	Persona Boundaries	Use archetype cards to co-create the VUI's persona and surface acceptable emotional and ethical boundaries, addressing older adults' concerns about data privacy and "false empathy" in AI systems [14].

**Figure 2: Structured flow of the co-design session.**

“It could be either... digital community as well... reminiscing about time playing *World of Warcraft*,” P7).

Visual, conversational, and multimodal triggers for reminiscence. Photographs were the most meaningful recall catalysts with *nearly all* describing them as conversation anchors that prompt storytelling rather than standalone artifacts (“Yes, old pictures always help with reminiscence... if you look at an old picture, it lets you tell a story or think about that time,” P7). Most participants also valued music, video, and voice recordings for evoking emotion and situating memories in time (“When I travel I buy a CD of local music... playing my Portuguese fado disc, I’m back in Portugal,” P11), while some emphasized sensory cues like food, smell, or objects (“I have like a mini altar... stones from Brazil... a stone from Machu Picchu,” P9). Across accounts, meaningful reminiscence emerged

through combinations of visual, auditory, and conversational cues (“All of these things, music, photograph, video conversations linked to those memories,” P6).

4.3 RQ3: Desired Functional Features for Reminiscence Systems

Core features. Most participants emphasized features that support co-creation, sharing, and legacy-building, valuing memories that could be transformed into artifacts accessible to family members and future generations. Most participants valued recording memories in their own voice and creating printable photo albums, emphasizing authenticity and materiality in legacy-building (“Must have would be create a printable photo album of my stories... just

having that collection immediately accessible,” P7). Most participants highlighted co-creation with family members for incorporating multiple perspectives or identifying people and places in photographs (“...let me and my family member add to the same memory from each from our own perspectives... History can be retold many ways,” P3).

Secondary and Optional Features. Secondary features enhanced engagement without being essential. Some participants valued music, media playback, or adaptive prompts tied to memories (“Like maybe if the grandmother starts talking about something, maybe a prompt from the screen can be, oh, do you want to call your daughter and talk to her about it...,” P2). A few participants were open to optional enhancements such as fictionalized stories, historical context, or interpretive overlays, with caution that these should not override personal narratives. Across discussions, participants emphasized balancing active facilitation with flexibility, supporting natural, conversational, and socially meaningful engagement (“...lead the conversation and help in a way that kind of probes and kind of gets deeper into the memory,” P4).

5 Discussion

Prior work on technological reminiscence systems has emphasized usability, structured prompting, multimedia access, and therapeutic protocols [13, 28]. Building on this work, we articulate design guidelines (DGs) that translate system-level insights into interactionally grounded guidance for voice-based reminiscence systems. First, systems should be designed as relational presences (DG1) rather than task-oriented tools, subsuming the need for intelligent facilitation and emotional sensitivity. Participants evaluated the *Memory Box* in terms of attentiveness, conversational pacing, and emotional safety, valuing systems that listen more than they speak, avoid directive or clinical questioning, and demonstrate conversational judgment through sensitivity to pauses, hesitation, and emotionally charged moments. Intelligence was associated not with factual correctness but with emotional attunement and restraint, extending prior work on empathetic yet non-intrusive VUI interaction to reminiscence [11]. Second, memory cues such as photos, videos, or music should function as conversational anchors (DG2) rather than primary content, contrasting with systems that foreground artifacts as core interventions [18, 22]. Meaningful reminiscence emerged through combinations of visual, auditory, and conversational cues, with no single trigger sufficient alone. Third, reminiscence systems should be socially situated and intergenerational (DG3), supporting co-creation, sharing, and legacy-building through recorded voices and tangible memory artifacts for family members rather than private archival storage alone. Finally, sustained engagement depends on continuity over time (DG4), including remembered preferences and growing conversational familiarity, supported by lightweight interaction that respects sensory and cognitive constraints through flexible pacing, easy pausing, and effortless re-entry. Together, these DGs reframe reminiscence systems as emotionally safe, socially embedded, and relationally continuous conversational experiences rather than mechanisms for structured content delivery.

6 Limitations and Future Work

Our findings should be interpreted in light of the small and relatively homogeneous sample, which consisted primarily of highly educated, English-speaking older adults in the United States; broader demographic, cultural, and socioeconomic diversity may surface different expectations, interaction styles, and concerns around voice-based reminiscence systems. Although our motivation foregrounded risks associated with LLM-based systems, such as hallucination, over-personalization, and misplaced trust, these risks were not deeply integrated into the empirical findings or design guidelines. Future work should more directly examine how such LLM-specific risks manifest in reminiscence contexts and how relational design principles can mitigate potential harms while preserving emotional attunement.

7 Conclusion

We co-designed *Memory Box* as a voice-based, multimodal system, which participants perceived not as a tool but as a relational presence evaluated by attentiveness, emotional safety, conversational continuity, and social fit. This reframing led to *MeBo*, a friendlier, persona-shaped companion emphasizing conversational sensitivity and appropriate emotional boundaries over clinical authority. Intelligence was understood as judgment and restraint, with memory cues as optional anchors for shared remembering. In future work, **we will operationalize these design guidelines by developing a functional MeBo prototype and conducting longitudinal studies to examine how such relational expectations and memory practices evolve over time.** Concerns around data privacy, control, and appropriate boundaries of memory storage and sharing that surfaced will guide design trade-offs in developing the *MeBo* prototype.

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